Trend Analysis: Reading the Historical Charts

> Part of the Functional Series. Before any forecast, the platform describes what has already happened. This document explains the measures behind the historical charts, what each one is good for, and the mistakes each invites. The data behind them is described in Data Sources.

1. The measures

Measure	What it answers	Where it appears	Watch out for
Level ($MtCO_2$)	How much is emitted?	Overview, Country Profile, Data Explorer	Large countries dominate any comparison of levels
Year-on-year change (%)	Is the country emitting more or less than last year?	Country Profile (YoY chart)	Noisy; one recession, pandemic or data revision swings a single year
Change since a base year (%, or index with base = 100)	How far has it moved from the starting point?	Historical Trends ("Top Movers" and "% Change Since 1990", with a Year control), Scenario Comparison	The answer depends on the base year chosen (§2)
Per capita (tCO_2 per person)	How much per resident?	Country Profile	Hides total scale: a small, rich country can be high per person and tiny in total
Share (% of a total)	Who accounts for how much?	Overview share section, gas-composition charts	A share can fall while the level rises, if others grow faster
Composition by decade (% by gas)	Which gases make up the total?	Historical Trends	Needs a source that covers all gases (PRIMAP-hist), so it starts later than the CO_2 series
Cumulative ($GtCO_2$)	How much has been emitted in total since a start year?	Overview map (cumulative view), Home's Top 10 ranking race, Climate Correlation	Warming follows the cumulative total, not any single year

Colour convention. Decreases in emissions are shown in green and increases in crimson, on every chart where a direction is judged. A *forecast* is shown in a neutral colour, because a forecast is a series, not a verdict.

2. The base-year problem

"Change since 1990" is a ratio, so its value depends on where 1990 sat in a country's own history. A country that had a bad year in 1990 looks better against it, and a country that had a good year looks worse. The platform therefore says which baseline each view uses:

View	Baseline	Why
Overview, Historical Trends, Scenario Comparison	1990 = 100	Matches the model training window (1990 onward) used across the platform
Country Profile trend	Full OWID history	1990 is shown as a marker, not as the start
All-gas relationship, gas composition	1970 onward	Earlier PRIMAP-hist values are reconstructions, not measurements
Long-run CO_2 relationship	1850–1900 reference	Pre-industrial context keeps early accumulation visible
Global, non-ML totals	Full available range	No model-consistency constraint applies

1990 has a second meaning the reader should know. It is the reference year of the Kyoto-era climate agreements, and for many countries it is a meaningful starting point. In some it is also distorted: German reunification and the collapse of the Soviet economy both occurred around it, which cut emissions in the affected countries for reasons unrelated to climate policy. A country's decline since 1990 is therefore not the same as the effect of its climate policy.

3. Which countries, and the aggregation choices

- **Featured (10) and Selected (up to 10)** drive the readable comparison charts. Past ten lines, a chart stops being readable.
- **Expanded (40)** is used for totals, the scenario treemap and model evaluation. These countries hold 92.2% of the 2024 global total, so an expanded-set total approximates the world, but it is not the world.
- **Non-sovereign entries** (regions, income groups, the European Union aggregates, international transport) are excluded from every country ranking. They would double count the countries they contain.

A global total from the World row and a total from summing countries differ by design: the World row includes international aviation and shipping, and the sum of countries does not.

4. A short reading guide

1. **Say what is plotted**: level, rate, index or share, and in which unit.
2. **Check the base year** before comparing two "% change" figures.
3. **Separate total from per capita**: they answer different questions and can point in opposite directions.
4. **Look at several years, not one.** The pandemic year 2020 sits inside most series and is a drop that reversed; a single point there is not a trend.
5. **Remember the scope**: territorial emissions, excluding emissions embodied in trade.

See also

- Forecasting: extending these series forward
- Limitations and Uncertainty

- Technical: EDA and Data Engineering
- Technical: Feature Engineering: the lag, rolling and percentage-change features computed from these series